

## Assignment No 28

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Domain : AI & ML

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Q1.

\*\*\* First 10 Rows:

|   | text                                              | emotions |
|---|---------------------------------------------------|----------|
| 0 | i didnt feel humiliated                           | sadness  |
| 1 | i can go from feeling so hopeless to so damned... | sadness  |
| 2 | im grabbing a minute to post i feel greedy wrong  | anger    |
| 3 | i am ever feeling nostalgic about the fireplac... | love     |
| 4 | i am feeling grouchy                              | anger    |
| 5 | ive been feeling a little burdened lately wasn... | sadness  |
| 6 | ive been taking or milligrams or times recomme... | surprise |
| 7 | i feel as confused about life as a teenager or... | fear     |
| 8 | i have been with petronas for years i feel tha... | joy      |
| 9 | i feel romantic too                               | love     |

Shape of Dataset:  
(16000, 2)

Missing Values:  
text 0  
emotions 0  
dtype: int64

Q2.

```
... Unique Emotion Labels:
['sadness' 'anger' 'love' 'surprise' 'fear' 'joy']

Label Mapping:
{'anger': np.int64(0), 'fear': np.int64(1), 'joy': np.int64(2), 'love': np.int64(3), 'sadness': np.int64(4), 'surprise': np.int64(5)}

Dataset with Encoded Labels:
```

|   | text                                              | emotions \ |
|---|---------------------------------------------------|------------|
| 0 | i didnt feel humiliated                           | sadness    |
| 1 | i can go from feeling so hopeless to so damned... | sadness    |
| 2 | im grabbing a minute to post i feel greedy wrong  | anger      |
| 3 | i am ever feeling nostalgic about the fireplac... | love       |
| 4 | i am feeling grouchy                              | anger      |
| 5 | ive been feeling a little burdened lately wasn... | sadness    |
| 6 | ive been taking or milligrams or times recomme... | surprise   |
| 7 | i feel as confused about life as a teenager or... | fear       |
| 8 | i have been with petronas for years i feel tha... | joy        |
| 9 | i feel romantic too                               | love       |

|   | emotion_encoded |
|---|-----------------|
| 0 | 4               |
| 1 | 4               |
| 2 | 0               |
| 3 | 3               |
| 4 | 0               |
| 5 | 4               |
| 6 | 5               |
| 7 | 1               |
| 8 | 2               |
| 9 | 3               |

Q3.

Before and After Lowercasing:

Memory usage: 278 MB

Sample 1

Before : i didnt feel humiliated

After : i didnt feel humiliated

-----

Sample 2

Before : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

After : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

-----

Sample 3

Before : im grabbing a minute to post i feel greedy wrong

After : im grabbing a minute to post i feel greedy wrong

-----

Sample 4

Before : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

After : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

-----

Sample 5

Before : i am feeling grouchy

After : i am feeling grouchy

-----

Q4.

Before and After Punctuation Removal:

Sample 1

Before : i didnt feel humiliated

After : i didnt feel humiliated

-----

Sample 2

Before : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

After : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

-----

Sample 3

Before : im grabbing a minute to post i feel greedy wrong

After : im grabbing a minute to post i feel greedy wrong

-----

Sample 4

Before : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

After : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

-----

Sample 5

Before : i am feeling grouchy

After : i am feeling grouchy

-----

Q5.

Before and After Removing Numbers:

Sample 1

Before : i didnt feel humiliated

After : i didnt feel humiliated

-----

Sample 2

Before : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

After : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

-----

Sample 3

Before : im grabbing a minute to post i feel greedy wrong

After : im grabbing a minute to post i feel greedy wrong

-----

Sample 4

Before : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

After : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

-----

Sample 5

Before : i am feeling grouchy

After : i am feeling grouchy

-----

Q6.

Before and After Removing Emojis & Special Characters:

Sample 1

Before : i didnt feel humiliated

After : i didnt feel humiliated

-----

Sample 2

Before : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

After : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i

-----

Sample 3

Before : im grabbing a minute to post i feel greedy wrong

After : im grabbing a minute to post i feel greedy wrong

-----

Sample 4

Before : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

After : i am ever feeling nostalgic about the fireplace i will know that it is still on the property

-----

Sample 5

Before : i am feeling grouchy

After : i am feeling grouchy

-----

Q7.

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
Before and After Removing Stopwords:

Sample 1
Before : i didnt feel humiliated
After  : didnt feel humiliated
-----

Sample 2
Before : i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and i
After  : go feeling hopeless damned hopeful around someone cares awake
-----

Sample 3
Before : im grabbing a minute to post i feel greedy wrong
After  : im grabbing minute post feel greedy wrong
-----

Sample 4
Before : i am ever feeling nostalgic about the fireplace i will know that it is still on the property
After  : ever feeling nostalgic fireplace know still property
-----

Sample 5
Before : i am feeling grouchy
After  : feeling grouchy
-----
```



Q8.

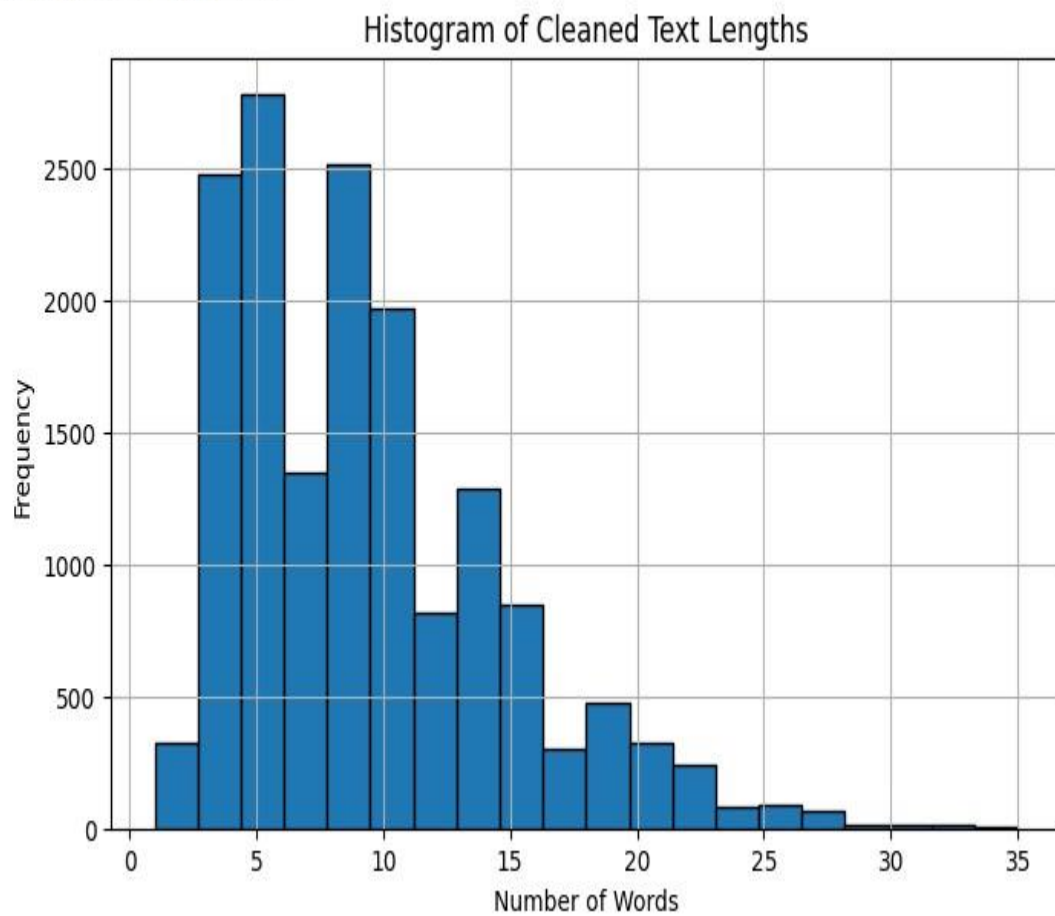
```
▼ ... [nltk_data] Downloading package stopwords to /root/nltk_data...
      [nltk_data]   Package stopwords is already up-to-date!

      original_text \
0          i didnt feel humiliated
1 i can go from feeling so hopeless to so damned...
2 im grabbing a minute to post i feel greedy wrong
3 i am ever feeling nostalgic about the fireplac...
4          i am feeling grouchy
5 ive been feeling a little burdened lately wasn...
6 ive been taking or milligrams or times recomme...
7 i feel as confused about life as a teenager or...
8 i have been with petronas for years i feel tha...
9          i feel romantic too

      cleaned_text
0          didnt feel humiliated
1 go feeling hopeless damned hopeful around some...
2 im grabbing minute post feel greedy wrong
3 ever feeling nostalgic fireplace know still pr...
4          feeling grouchy
5 ive feeling little burdened lately wasnt sure
6 ive taking milligrams times recommended amount...
7 feel confused life teenager jaded year old man
8 petronas years feel petronas performed well ma...
9          feel romantic
```

Q9.

Average Text Length : 9.3530625  
... Minimum Text Length : 1  
Maximum Text Length : 35



Q10.

```
*** [nlTK_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
First 5 Rows:
      text emotions
0      i didnt feel humiliated sadness
1 i can go from feeling so hopeless to so damned... sadness
2 im grabbing a minute to post i feel greedy wrong anger
3 i am ever feeling nostalgic about the fireplac... love
4      i am feeling grouchy anger

Emotion Label Mapping:
{'anger': np.int64(0), 'fear': np.int64(1), 'joy': np.int64(2), 'love': np.int64(3), 'sadness': np.int64(4),

Cleaned dataset saved successfully as 'cleaned_emotions.csv'

Emotion Value Counts:
emotions
joy      5362
sadness  4666
anger    2159
fear     1937
love     1304
surprise  572
Name: count, dtype: int64

Sample of Cleaned Dataset:
      text \
0      i didnt feel humiliated
```

```

-
1 i can go from feeling so hopeless to so damned...
2 im grabbing a minute to post i feel greedy wrong
3 i am ever feeling nostalgic about the fireplac...
4 i am feeling grouchy
5 ive been feeling a little burdened lately wasn...
6 ive been taking or milligrams or times recomme...
7 i feel as confused about life as a teenager or...
8 i have been with petronas for years i feel tha...
9 i feel romantic too

```

|   | cleaned_text                                      | emotions \ |
|---|---------------------------------------------------|------------|
| 0 | didnt feel humiliated                             | sadness    |
| 1 | go feeling hopeless damned hopeful around some... | sadness    |
| 2 | im grabbing minute post feel greedy wrong         | anger      |
| 3 | ever feeling nostalgic fireplace know still pr... | love       |
| 4 | feeling grouchy                                   | anger      |
| 5 | ive feeling little burdened lately wasnt sure     | sadness    |
| 6 | ive taking milligrams times recommended amount... | surprise   |
| 7 | feel confused life teenager jaded year old man    | fear       |
| 8 | petronas years feel petronas performed well ma... | joy        |
| 9 | feel romantic                                     | love       |

|   | emotion_encoded |
|---|-----------------|
| 0 | 4               |
| 1 | 4               |
| 2 | 0               |
| 3 | 3               |
| 4 | 0               |
| 5 | 4               |
| 6 | 1               |